

Requirements:

Transportation company is seeking to gain deeper insights into its operational and financial performance through data analysis. With a focus on optimizing efficiency, reducing costs, and enhancing service delivery, the company aims to analyze key data points across various aspects of its logistics and transportation services.

Analysis Questions:

1. Operational Efficiency

- What is the average shipping duration (ShipDays) by origin to destination state?
- Which routes (Origin city-Destination city pairs) have the longest and shortest average delivery times?
- How many trips are completed per shipper, and how do their average delivery times compare?

2. Cost & Revenue Analysis

- What is the overall profit margin per trip (Revenue - ShippingCost)?
- Which trips or routes generate the highest revenue?
- What is the cost per mile and revenue per mile for different trip?

3. Capacity & Utilization

- What is the average truck capacity?
- Average capacity by trip type?
- Calculate load factor?

4. Customer Insights

- Which customers have the most shipments?
- Find top 10 customer by total profit?
- Which customers are associated with the longest shipping times?

5. Geographic Insights

- Which cities are the most frequent origins and destinations?
- How do revenue and cost vary by origin state?
- What are the Average Revenue by state to state transport?

6. Trip Trends & Volume

- How many trips are completed each month?
- Are there any seasonal trends in shipping volume?
- What is the trend of average shipping cost over time?

7. Performance Monitoring

- Which trips exceed expected delivery time (Average ShipDays) compared to actual (DeliveryDate - ShipDate)?
- How many checkpoints are passed per trip, and how do they correlate with delays?
- Which shipper is the most reliable in terms of on-time delivery?